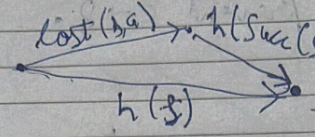
How to know which heuristic is better?

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① Consistent heuristic



$cost(s,a) + h(Succ(s,a)) \geq h(s)$

$cost(s,a) + h(Succ(s,a)) - h(s) \geq 0$

② Admissible

Local Search

* Modern day problem
  - Local restaurants near Jaipur railway station (finding something within specific geographical area)

Note: unlike systematic search, local search sol^n is not the path but the state within a sol^n space.

Path vs state optimization (in local search)

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Local search & optimization

- ↳ keep track of single state
- ↳ move only to neighbouring state (succ° fn)
- ↳ ignore paths

⇒ **Adv:** Little memory

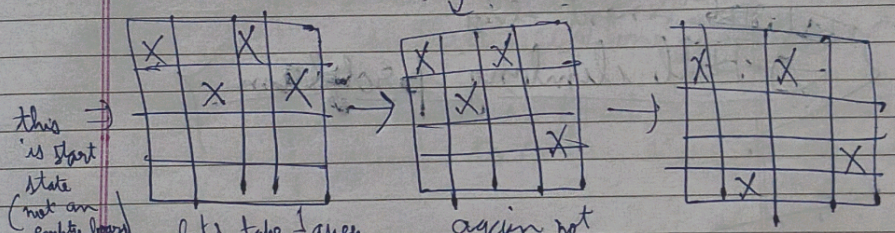
① Can often find 'sol' in large state space

Eg - **N Queens problem**

↳ CSP vs COP?

- any sol" satisfying constraints
- ↳ min moves

Consider New changed problem



this ⇒ is start state (not an empty board) we need to change such that it satisfies N queens.

lets take 1 queen and have it secure (not attacked by anyone)

again not solved, so take 1 more and secure

'n-1' positions (in column)

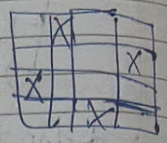
then check for valid

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Trivial sol^n: Case 1: Random sample
Case 2: Random walk

Asymptotically complete!
(as long as connected)



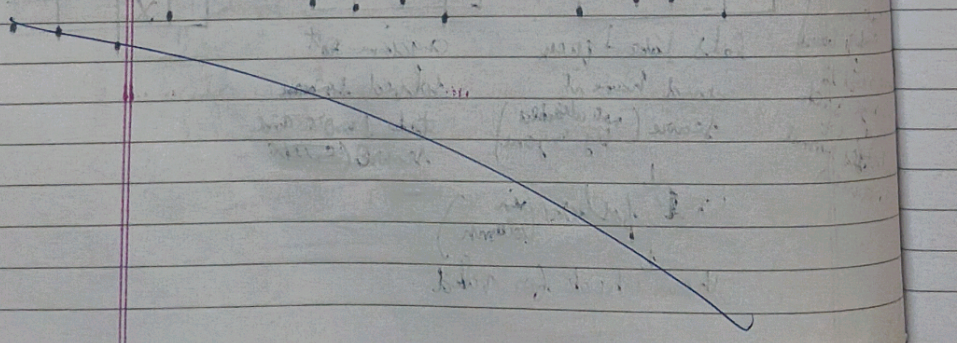
$O(n^n)$

Simple fix: Greedy local search

1. Jump to neighbour having best objective function
2. Repeat
3. Needs termination criteria
[none of the neighbour has objective function better than me]

Hill climbing problem

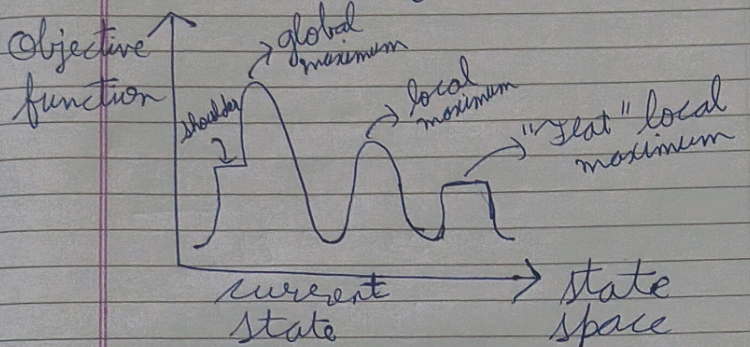
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Hill Climbing Problem



Drawback: (1) stuck in local optima.

(2) Depends on start state (S_0).

Fix: * stochastic HC

Reduces complexity

- Whenever in local optima, randomly sample a neighbour [random walk with p]
- Whenever in local optima, randomly jump to other state [random sample with p]
- p increases over time